10 Collections 2 – Dictionaries

Dictionaries Team 1

Create a program called **dictionaries_team_1**. The program should first create a dictionary that holds each team member's name (as the key) and role in the team (as the value). Then the program should print the dictionary as shown in the example output.

The team members are the following:

John	software developer
Ann	project manager
Susan	software developer
Jill	lead developer

Example output:

```
{'John': 'software developer', 'Ann': 'project manager', 'Susan': 'software developer', 'Jill': 'lead developer'}
```

Dictionaries Team 2

Create a program called **dictionaries_team_2**. Copy/paste the dictionary from the *Dictionaries Team 1* exercise to this program. The program should print the team members as shown in the example output.

Example output:

```
John Ann Susan Jill
Ann Jill John Susan
```

Dictionaries Team 3

Create a program called **dictionaries_team_3**. Copy/paste the dictionary from the *Dictionaries Team 1* exercise to this program. The program should ask the user to enter a team member's name and print the team member's role in the team. If the name is not in the dictionary, the program should print the name and "is not in the team". The program should be repeating these steps until the user enters "quit" when the program asks the user to enter a name.

Example output:

```
Enter name (quit ends): Frank
Frank is not in the team

Enter name (quit ends): Ann
Ann is a project manager

Enter name (quit ends): Susan
Susan is a software developer

Enter name (quit ends): quit
```

Dictionaries Translator

Create a program called **dictionaries_translator**. The program should first input Finnish and English words from the user until the user enters "quit". The program should save the inputted words to a dictionary. The words should be saved in lowercase.

Next, the program should input an English word from the user and print the corresponding Finnish word. If the English word does not exist in the dictionary, the program should print "Unknown word". The program should keep repeating these steps until the user enters "quit" when the program asks the user to enter an English word.

Example output:

```
=== Creating a dictionary ===
Enter english word (quit ends): CAT
Enter finnish word: kisSA
Enter english word (quit ends): dog
Enter finnish word: koira
Enter english word (quit ends): quit

=== Using a dictionary ===
Enter english word (quit ends): caT
kissa

Enter english word (quit ends): python
Unknown word

Enter english word (quit ends): Dog
koira

Enter english word (quit ends): quit
```

Dictionaries Team 4

Create a program called **dictionaries_team_4** Copy/paste the dictionary from the *Dictionaries Team 1* exercise to this program.

The program should ask the user to enter name and role. If the name is not in the dictionary, the program should add the inputted data to the dictionary. If the name is already in the dictionary, the program should update the team member's role with the inputted role. The program should keep repeating these steps until the user enters "quit" when the program asks the user to enter a name.

Finally, the program should print the team members' names and roles as shown in the example output.

Please ***set the output width to 8*** when you print team members' names.

Example output:

```
Enter name (quit ends): Mike
Enter role: software developer

Enter name (quit ends): Rosanna
Enter role: ux designer

Enter name (quit ends): quit

Ann     project manager
Jill    lead developer
John    software developer
Mike    software developer
Rosanna ux designer
Susan   software developer
```

Dictionaries Team 5

Create a program called **dictionaries_team_5** Copy/paste the dictionary from the *Dictionaries Team 1* exercise to this program.

The program should print team members' roles as shown in the example output. Please notice that each role is printed only once.

Example output:

```
lead developer
project manager
software developer
```

Dictionaries Count Letters

Create a program called **dictionaries_count_letters**. The program should first fetch the content of a text from the given URL and save it as single a string. This we can do with the code below.

```
from urllib.request import urlopen

url = "https://www.mit.edu/~ecprice/wordlist.10000"
content = urlopen(url).read().decode("UTF-8")
```

Then the program should count the letters in the string and display the result as shown in the example output below.

In this exercise,

- you should ***use a dictionary*** to hold the count of occurrences of each distinct letter in the string
- you should count letters in a case-insensitive way
- you should not display a letter if it does not appear in the string.

Example output:

```
a: 5378
b: 1141
c: 3025
d: 2507
e: 7601
f: 927
g: 1717
h: 1429
i: 5461
j: 183
k: 592
l: 3231
m: 1912
n: 4822
o: 4252
p: 2027
q: 123
r: 4860
s: 5085
t: 4760
u: 1939
v: 849
w: 632
x: 264
y: 1027
z: 136
```

Dictionaries Uni Database 1 (*This is a grade level 4-5 exercise*)

Create a program called **dictionaries_uni_database_1**. The program should save facts about degree programs and their courses in a dictionary. The facts per each degree program include the following:

- degree program name (the key in the dictionary)
- courses, which the degree program provides (the value in the dictionary – *another collection*)

NB! Here we need some planning. Please, think carefully what kinds of collections you want to use in your program.

First, copy/paste the *main function* below to your program. You should not modify the main function. Then write the other functions.

```
def main():
    db = {}

    insert_degree_program(db, "ITBBA")
    insert_degree_program(db, "EXPER")

    insert_course(db, "ITBBA", ("Python Programming", 5))
    insert_course(db, "ITBBA", ("Time Management", 3))
    insert_course(db, "EXPER", ("Creative Hospitality and Tourism", 5))
    insert_course(db, "EXPER", ("Managing Dynamic Restaurant Business", 10))

    print_degree_program(db, "ITBBA")
    print_degree_program(db, "EXPER")

main()
```

The **insert_degree_program** function

- should print the **degree program name** and " **is already in the database**" if the degree program name already exists in the dictionary.

The **insert_course** function

- should print "**Unknown degree program:** " and the **degree program name** if the degree program name does not exist in the dictionary.
- should print the **course name** and " **is already in the database**" if the course already exists in the degree program.

When you run the program, it should print exactly what is shown in the example output below.

Example output:

```
ITBBA (2 courses)
  Python Programming, 5 credits
  Time Management, 3 credits
  Total credits: 8
EXPER (2 courses)
  Creative Hospitality and Tourism, 5 credits
  Managing Dynamic Restaurant Business, 10 credits
  Total credits: 15
```

Dictionaries Uni Database 2 (*This is a grade level 4-5 exercise*)

Create a program called **dictionaries_uni_database_2**. Please follow the instruction below.

1. Save a copy of **dictionaries_uni_database_1.py** as **dictionaries_uni_database_2.py**
2. Replace the old *main function* with the code below. Do should not modify the new main function.

```
def main():
    db = {}

    insert_degree_program(db, "ITBBA")
    insert_degree_program(db, "EXPER")

    insert_course(db, "ITBBA", ("Python Programming", 5))
    insert_course(db, "ITBBA", ("Time Management", 3))
    insert_course(db, "EXPER", ("Creative Hospitality and Tourism", 5))
    insert_course(db, "EXPER", ("Managing Dynamic Restaurant Business", 10))

    print_degree_program(db, "ITBBA")
    print_degree_program(db, "EXPER")
    print()

    remove_course(db, "ITBBA", "Python Programming")
    print_degree_program(db, "ITBBA")
    print()

    # Testing error handling
    insert_degree_program(db, "ITBBA")
    insert_course(db, "ITBBA", ("Time Management", 3))

    print_degree_program(db, "LOBBY")
    remove_course(db, "COMPU", "Surfing")
    remove_course(db, "ITBBA", "Surfing")

main()
```

3. Write the **remove_course** function. The function

- should remove a course from a degree program.
- should print "**Unknown degree program:** " and the **degree program name** if the degree program name does not exist in the dictionary.
- should print "**Unknown course:** " and the **course name** if the course does not exist in the degree program.

When you run the program, it should print exactly what is shown in the example output below.

Example output:

```
ITBBA (2 courses)
  Python Programming, 5 credits
  Time Management, 3 credits
  Total credits: 8
EXPER (2 courses)
  Creative Hospitality and Tourism, 5 credits
  Managing Dynamic Restaurant Business, 10 credits
  Total credits: 15

ITBBA (1 course)
  Time Management, 3 credits
  Total credits: 3

ITBBA is already in the database
Time Management is already in the database
Unknown degree program: LOBBY
Unknown degree program: COMPU
Unknown course: Surfing
```

Dictionaries Transpose (*This is a grade level 4-5 exercise*)

Create a program called **dictionaries_transpose**. The program should have a function named **transpose_dictionary**. The function should take a dictionary as argument and transpose the dictionary so that it swaps the key and the value in each dictionary entry.

First, copy/paste the *main function* below to your program. Then write the **transpose_dictionary** function.

```
def main():
    vocabulary = {"cat": "kissa", "dog": "koira", "bird": "lintu"}
    print(vocabulary)

    transpose_dictionary(vocabulary)
    print(vocabulary)

main()
```

Example output:

```
{'cat': 'kissa', 'dog': 'koira', 'bird': 'lintu'}
{'kissa': 'cat', 'koira': 'dog', 'lintu': 'bird'}
```